Homework 4 Solutions

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Section 2.1

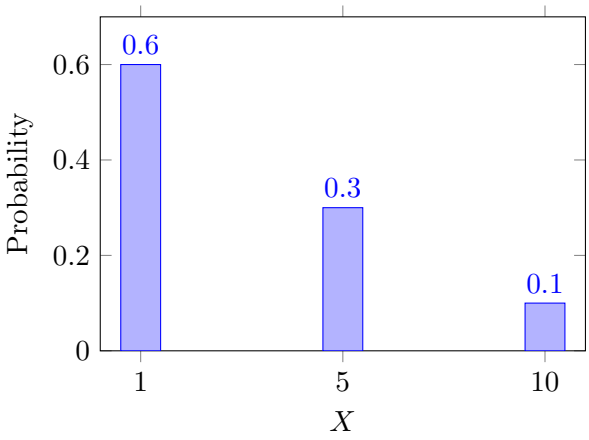
2.1-2

We have 6 white, 3 red, 1 blue chips. Total = 10.

$$P(X=1)=\tfrac{6}{10}=0.6, \quad P(X=5)=\tfrac{3}{10}=0.3, \quad P(X=10)=\tfrac{1}{10}=0.1$$

This is the pmf.

Bar Graph:



2.1-4

The pmf is

$$f(x)=\log_{10}\left(\frac{x+1}{x}\right), \quad x=1,2,\ldots,9.$$

(a) To check it's a pmf:

$$\sum_{x=1}^{9} f(x)=\sum_{x=1}^{9}\log_{10}\left(\frac{x+1}{x}\right)=\log_{10}\left(\frac{10}{1}\right)=1.$$

(b) The cdf is

$$F(x) = P(X \le x) = \sum_{k=1}^{x} f(k) = \log_{10}(x+1), \quad x = 1, \ldots, 9.$$

Section 2.2

2.2-2

The pmf is $f(x) = \frac{(|x|+1)^2}{9}$, $x = -1, 0, 1$.

$$f(-1) = \tfrac{4}{9}, \quad f(0) = \tfrac{1}{9}, \quad f(1) = \tfrac{4}{9}.$$

$$E(X) = (-1)\tfrac{4}{9} + 0 + \tfrac{4}{9} = 0,$$
$$E(X^2) = (1)\tfrac{4}{9} + 0 + (1)\tfrac{4}{9} = \tfrac{8}{9},$$
$$E(3X^2 - 2X + 4) = 3E(X^2) - 2E(X) + 4 = 3 \cdot \tfrac{8}{9} + 0 + 4 = \tfrac{20}{3}.$$

2.2-4

We need c. Since probabilities must sum to 1:

$$0.9 + \sum_{x=1}^{6} \frac{c}{x} = 1 \quad \Rightarrow \quad c = \frac{0.1}{49/20} = \frac{2}{49}.$$

The company pays $(x-1)$ whenever $x \ge 1$.

$$E(\text{payment}) = \sum_{x=1}^{6} (x-1) \cdot \frac{c}{x} = c \sum_{x=1}^{6} \frac{x-1}{x}.$$

The sum is $\frac{71}{20}$, so

$$E(\text{payment}) = \frac{2}{49} \cdot \frac{71}{20} = \frac{71}{490} \approx 0.145.$$

2.2-5

The pmf is $f(x) = \frac{5-x}{10}$, $x = 1, 2, 3, 4$. Payments: - $x = 1$: \$200 - $x = 2$: \$400 - $x = 3$: \$500 - $x = 4$: \$600

$$E(\text{payment}) = \sum_{x=1}^{4} (\text{payment}) \cdot f(x) = 200 \cdot 0.4 + 400 \cdot 0.3 + 500 \cdot 0.2 + 600 \cdot 0.1 = 360.$$

Section 2.3

2.3-1

(a) $X = 5$ with prob 1. Then $\mu = 5$, variance $= 0$. Skewness is usually taken as 0 (degenerate case).

(b) Uniform on $\{1, 2, 3, 4, 5\}$. Mean $= 3$, variance $= 2$, skewness $= 0$.

(c) Uniform on $\{3, 5, 7, 9, 11\}$. Mean $= 7$, variance $= 8$, skewness $= 0$.

(d) $f(x) = x/6$, $x = 1, 2, 3$. Then

$$E(X) = \tfrac{7}{3}, \quad E(X^2) = 6, \quad \mathrm{Var}(X) = 6 - (7/3)^2 = 5/9.$$

Third central moment $= -7/27$, so

$$\gamma_1 = \frac{-7/27}{(5/9)^{3/2}} = -\frac{7}{5\sqrt{5}}.$$

2.3-11

MGF: $M(t) = \frac{2}{5}e^t + \frac{1}{5}e^{2t} + \frac{2}{5}e^{3t}$. So

$$P(X = 1) = \tfrac{2}{5}, \quad P(X = 2) = \tfrac{1}{5}, \quad P(X = 3) = \tfrac{2}{5}.$$

$$E(X) = 1 \cdot \tfrac{2}{5} + 2 \cdot \tfrac{1}{5} + 3 \cdot \tfrac{2}{5} = 2,$$
$$E(X^2) = 1^2 \cdot \tfrac{2}{5} + 2^2 \cdot \tfrac{1}{5} + 3^2 \cdot \tfrac{2}{5} = \tfrac{24}{5},$$
$$\mathrm{Var}(X) = \tfrac{24}{5} - 2^2 = \tfrac{4}{5}.$$